INSTITUTE FOR ADVANCED COMPUTING AND SOFTWARE DEVELOPMENT, AKURDI, PUNE

"Fashion Fusion"

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Submitted By:

Group No: 49

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ABSTRACT

Ecommerce Clothing Website is a comprehensive platform designed to integrate online clothing with e-commerce functionalities. It allows businesses to efficiently track and manage their orders while providing customers with seamless online shopping experience. The platform uses technologies like React and Redux for a responsive front-end, and Spring Boot for a robust backend API. Security is enhanced through Spring Security for authentication and authorization, and payments.

Ecommerce Clothing website is a robust and scalable application designed to streamline orders management and facilitate seamless e-commerce operations. The project aims to address the challenges faced by retailers and wholesalers in managing their product orders and online sales through an integrated platform built with React for the frontend and Spring Boot for the backend.

ACKNOWLEDGEMENT

We take this occasion to thank God, almighty for blessing us with his grace and taking our endeavor to a successful culmination. We extend my heartfelt thanks to our esteemed guide, **Mrs. Megha Mane.** for providing us with the right guidance and advice at the crucial juncture and showing us the right way. We sincerely thank our respected Centre Co-Ordinator, **Mr. Rohit Puranik**, for allowing us to use the available facilities. We would also like to thank the other faculty members at this occasion. Last but not least, we would like to thank my friends and family for the support and encouragement they have given me during our work.

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1. INTRODUCTION

The evolution of digital commerce has significantly transformed the fashion industry, offering customers a seamless and convenient shopping experience. An e-commerce men's clothing website serves as an online platform where users can browse, select, and purchase a wide range of men's fashion items, including casual wear, formal wear and over-sized T-shirts.

This website is designed to cater to modern consumers who prefer hassle-free online shopping, providing them with a variety of styles, sizes, and brands at their fingertips. With the increasing demand for personalized shopping experiences, the platform incorporates features such as product recommendations, size guides, secure payment options, and order tracking.

By leveraging cutting-edge web development technologies and secure backend infrastructure, this platform aims to offer an efficient, engaging, and secure shopping experience tailored to men's fashion needs. Whether it's casual wear, business attire, or seasonal fashion, this e-commerce website strives to be a one-stop destination for modern men's clothing

In the **Ecommerce Clothing Website** system, both **User, Admin** and **Delivery Person** roles play crucial roles in the management and operation of the platform. Each role comes with specific responsibilities, permissions, and access levels to ensure that the system operates smoothly and securely.

1. Admin Role

The admin role is the most powerful in the Ecommerce Clothing website system, having full control over the platform's functionalities. Admins are typically business owners, managers, or IT personnel who need comprehensive access to manage the orders, users, and overall system settings.

❖ Responsibilities:

- **User Management:**
  - Create, manage, and delete user accounts.

- Assign roles and permissions to users.
- Monitor user activities and ensure compliance with company policies.

- **Orders Management:**
  - Add, update, or delete product information.
  - Monitor stock levels and ensure that orders data is up-to-date.
  - Set reorder levels for products to automate stock replenishment.

- **Order Management:**
  - View, process, and manage customer orders.
  - Oversee order fulfillment, including shipping and delivery processes.
  - Handle returns and exchanges.

- **Payment Management:**
  - Oversee and manage payments.

- **System Configuration:**
  - Configure system settings, including tax rates, discount rules, and shipping options.
  - Manage and update platform content, such as banners, promotions, and announcements.

- **Reporting & Analytics:**
  - Generate and analyse reports on sales, orders levels, user activities, and financial transactions.
  - Use data to make informed business decisions.

- **Security Management:**
  - Implement and enforce security policies using Spring Security.
  - Monitor security logs and respond to potential security breaches.

Access Level:

- Full access to all system features and data.
- Ability to override settings and actions performed by users.
- Access to sensitive financial and personal information of users and customers.

2. User Role

Users in the Orders Clothes shopping system typically refer to customers or employees with limited access. Their access level is more restricted compared to Admins, focusing on tasks related to orders tracking, order placement, or basic system interactions.

❖ Responsibilities:

- **Order Placement (for customers):**
  - Browse products, add items to the cart, and place orders.
  - Track order status and view order history.

- **Profile Management:**
  - Update personal information such as name, address, and contact details.
  - Manage account settings, including password changes.

- **Payments:**
  - Make payments for orders via the integrated payment gateway.

- **Access Level:**
  - Limited access based on assigned permissions.
  - Can view and interact with specific features (e.g., order placement, product browsing) but cannot alter system settings.
  - No access to sensitive system configurations, financial data, or user management functionalities.

Role-Based Access Control (RBAC)

The system implements Role-Based Access Control (RBAC) using **Spring Security**. This ensures that users only have access to the functionalities they need based on their roles. For instance:

- Admins have unrestricted access to all resources and can manage both the system and its users.
- Users have limited access, often restricted to viewing and interacting with orders items, placing orders, and managing their profiles.

Security Considerations

Both roles are secured through robust authentication mechanisms provided by Spring Security. This includes:

- **Authentication:** Verifying the identity of users before granting access to the system.
- **Authorization:** Ensuring that users can only access the resources permitted by their role.
- **Session Management:** Secure handling of user sessions to prevent unauthorized access.

It is role-based structure ensures that tasks are appropriately delegated and that the system remains secure and efficient. Admins can focus on high-level management and strategic decisions, while users interact with the system to fulfil their specific roles without compromising security.

1.1 Purpose

The purpose of the ecommerce clothing website is to provide a comprehensive and efficient platform for managing and selling clothes in a retail environment. The system is designed to streamline various business operations related to orders management, order processing, and customer interactions, ultimately enhancing the overall productivity and profitability of the business.

1.2 Scope

The e-commerce men’s clothing website aims to provide a seamless, secure, and user-friendly online shopping experience, catering specifically to men's fashion needs. With the integration of Spring Boot, Spring Security, React, and MySQL, the platform ensures efficient product management, secure transactions, and a dynamic user interface. The website supports user authentication, product browsing, cart management, order placement, and secure payment processing, along with an admin panel for orders and order management. The platform is designed for scalability, allowing future enhancements such as AI-driven recommendations, multi-vendor support, and mobile app integration. Given the continuous growth of the online fashion market, this project has significant potential to expand its reach, attract a broader customer base, and improve customer engagement through personalized shopping experiences and data-driven insights.

1.3 Objective of Orders Clothes shopping

The primary objective of this project is to develop a secure, scalable, and user-friendly Men's Clothing E-Commerce Website using Spring Boot, Spring Security, React, and MySQL. This platform aims to provide a seamless shopping experience for customers while ensuring efficient management for administrators.

1. Develop a Dynamic & Responsive Shopping Platform
2. Facilitate Efficient Product Management & Browsing
3. Enable a Seamless Shopping Experience
4. Robust Security and Access Control
5. Scalable and Flexible Architecture
6. Comprehensive Reporting and Analytics
7. Cost Optimization
8. Enhanced Customer Experience
9. Compliance and Audit Readiness
10. Support for Growth and Expansion

1.4 Functionalities Provided by Ecommerce Men's Clothing Website

1. User Management

- User Registration and Login:
    - Users can create accounts, log in, and manage their profiles.

   - Secure authentication and authorization using Spring Security.
- Role-Based Access Control:
   - Different user roles (e.g., Admin, Delivery Person, Customer) with specific permissions.
   - Admins can create and manage these roles and assign them to users.
- Profile Management:
   - Users can update their personal information, such as name, email, and contact details.
   - Password management, including options for changing and resetting passwords.

2. Orders Management

- Product CatLog Management:
   - Admins can add, update, and delete products in the orders.
   - Manage product details such as name, description, price, SKU, category, and images.
- Stock Level Monitoring:
   - Real-time tracking of orders levels for each product.
   - Automatic notifications for low stock levels to ensure timely reordering.
- Orders Adjustment:
   - Manual adjustments to orders levels to account for discrepancies, damage, or loss.
- Automated Reordering:
   - Automatic generation of purchase orders or alerts when orders falls below a predefined threshold.

3. Sales Management

- Order Processing:
   - Customers can place orders, and the system automatically processes them, updating orders levels accordingly.
   - Order confirmation and status updates are sent to customers.
- Payment Integration:

Supports multiple payment methods like (credit and debit cards).

- Order Tracking:
   - Customers can track the status of their orders, from processing to delivery.

4. Customer Management

- Customer Profiles:
   - Maintain detailed profiles for each customer, including order history, preferences, and contact information.
- Loyalty Programs and Discounts:

   - Implementation of loyalty programs or discounts to reward frequent customers and encourage repeat business.

5. Reporting and Analytics

- Sales Reporting:

  Detailed reports on sales performance, including revenue, profit margins, and top-selling products.

- Orders Reporting:

  Reports on orders levels, turnover rates, and stock valuation.

- Customer Analytics:

  Analysis of customer behavior, purchase patterns, and demographics to inform marketing strategies.

- Financial Reporting:

  Overview of financial metrics, including sales revenue, expenses, and profitability.

6. Security and Compliance

- Data Encryption:

  Encryption of sensitive data such as payment information and personal details to protect against unauthorized access.

- Audit Trails:

  Logging of significant actions within the system, such as orders changes, order processing, and user management for accountability and compliance.

- Compliance with Legal Requirements:

Ensuring that the system complies with relevant legal and regulatory requirements, such as data protection laws (e.g., GDPR).

7. Customer Experience Enhancement

- Search and Filtering:

  Advanced search and filtering options for customers to find products based on various criteria such as category, price, or brand.

- Wishlist and Shopping Cart:

  Features allowing customers to save products for future purchases or add them to their shopping cart.

- Responsive Design:

  A user interface that adapts to different devices, ensuring a seamless experience on desktops, tablets, and smartphones.

8. Integration and Extensibility

- API Integration:

  RESTful APIs for external systems to interact with the Men’s clothing website platform, enabling data exchange with other applications.

- Third-Party Integrations:

  Integration with third-party services such as accounting software, CRM systems, or e-commerce platforms.

2. SOFTWARE REQUIREMENT SPECIFICATION

The functional requirements for Ecommerce Clothing website outline the specific features and capabilities that the system must provide to meet the needs of its users. These requirements are essential for guiding the development process and ensuring that the final product aligns with the business objectives.

2.1 Functional Requirements for Ecommerce Men's Clothing Website

1. User Management

- **User Registration:**
  The system allow new users to create an account by providing personal details, such as name, email, and password.

- **User Authentication:**
  The system shall authenticate users during login using their registered email and password.

- **Role-Based Access Control:**
  The system shall support role-based access, where different users (Admin, Employee, Customer) have different permissions.

- **Profile Management:**
  Users shall be able to view and update their profiles, including personal details and passwords.

2. Orders Management

- **Product Management:**
  The system shall allow admins to add, update, and delete products from the orders.

  The system shall store product details such as name, description, SKU, price, category, and images.

- **Stock Level Monitoring:**

  The system should monitor stock levels in real time and display current orders levels.
- **Orders Adjustment:**

  The system shall allow manual adjustments to orders levels to account for discrepancies or damage.
- **Reordering:**

  The system shall automate reordering by generating purchase orders when stock levels are low.

3. Sales Management

- **Order Placement:**

  Customers shall be able to place orders for products from the orders.
- **Order Processing:**

  The system shall process orders, updating orders levels accordingly.
- **Payment Processing:**

  The system can handle online payments securely.

  The system shall support various payment methods, including credit/debit cards.
- **Order Tracking:**

  Customers shall be able to track the status of their orders from processing to delivery.

4. Customer Management

- **Customer Profiles:**

  The system shall maintain profiles for each customer, including order history and personal details.
- **Order History:**

  Customers shall be able to view their order history.
- **Loyalty Programs:**

  The system shall support the creation and management of loyalty programs for customers.

5. Reporting and Analytics

- **Sales Reports:**
  The system shall generate reports on sales performance, including revenue, profit margins, and top-selling products.

- **Orders Reports:**
  The system shall generate reports on orders levels, turnover rates, and stock valuation.
- **Customer Analytics:**
  The system shall provide insights into customer behavior, including purchase patterns and demographics.

6. Security

- **Data Encryption:**
  The system shall encrypt sensitive data, such as payment information and personal details.

- **Authentication and Authorization:**
  The system shall enforce strong authentication mechanisms and ensure that users have access only to the functionalities allowed by their roles.

2.2 Non-Functional Requirements for Orders Ecommerce men's Clothing website

1. Performance

- Response Time:

  The system shall respond to user actions within 2 seconds under normal operating conditions.
- Scalability:

  The system shall handle an increasing number of users and transactions without performance degradation. It should support at least 10,000 concurrent users.
- Throughput:

  The system shall process at least 100 transactions per second during peak usage times.

2. Reliability

- Availability:

  The system shall have an uptime of 99.9% over a 12-month period, ensuring high availability for users.
- Fault Tolerance:

  The system shall continue to operate in the event of hardware or software failures, with minimal disruption to users.
- Error Handling:

  The system shall gracefully handle errors and provide meaningful feedback to users when issues occur.

3. Usability

- User Interface:

  The system shall have a user-friendly interface that is easy to navigate, with clear instructions and minimal learning curve.

4. Maintainability

- **Modularity:**

  The system shall be designed in a modular fashion, allowing for easy updates and enhancements to individual components without affecting the entire system.

- **Code Quality:**

  The system shall follow coding best practices, with well-documented, clean, and maintainable code.

- **Testing:**

  The system shall undergo thorough testing, including unit tests, integration tests, and user acceptance tests, to ensure quality and stability.

❖ Other Requirements:

Hardware and Network Interfaces:

Back-end Server Configuration:

- Intel Core I5 Processor
- 8 GB RAM

Front-end Client Configuration:

- AMD RYZEN 5 Processor
- 128 MB SDRAM
- 10 GB Hard Disk Drive
- 104 Keys Keyboard
- PS2 Mouse with pad

Software Interfaces:

Software configuration for back-end Services:

- Java EE version JDK 17
- Spring Boot, JPA, Spring Authentication
- MySQL version 8.0.39
- STS version 4.0

Software configuration for front-end Services:

- ReactJS, Redux
- HTML, CSS, JS
- VS Code

❖ Future Scope:

- AI Powered Personalization & Recommendations.
- Augmented Reality (AR) & Virtual Try-On.
- Social Commerce and Influencer Marketing.
- Subscription & Membership Models.

3. DIAGRAMS

3.1 Entity Relationship Diagram:

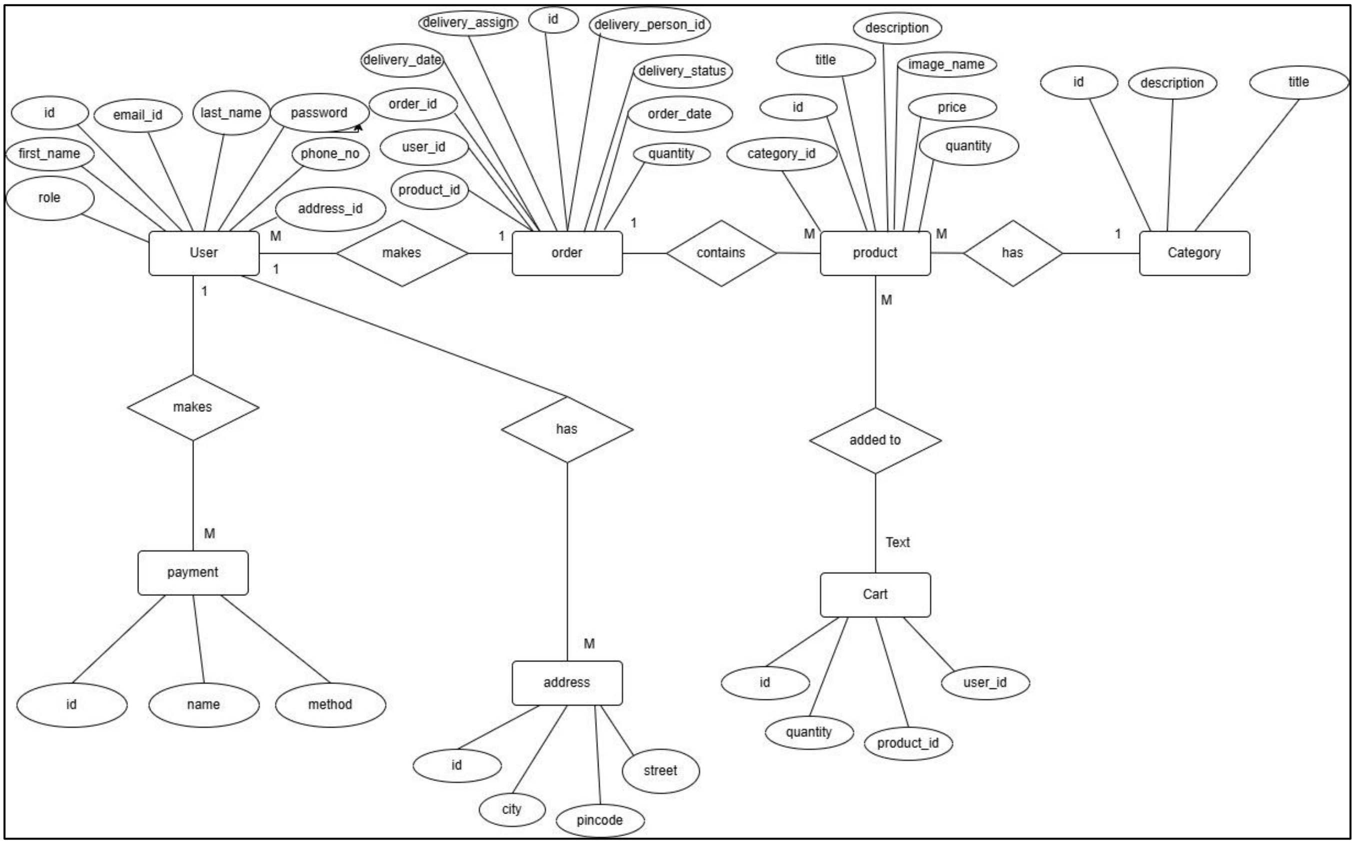


Fig. ER Diagram

3.2 Use Case Diagram:

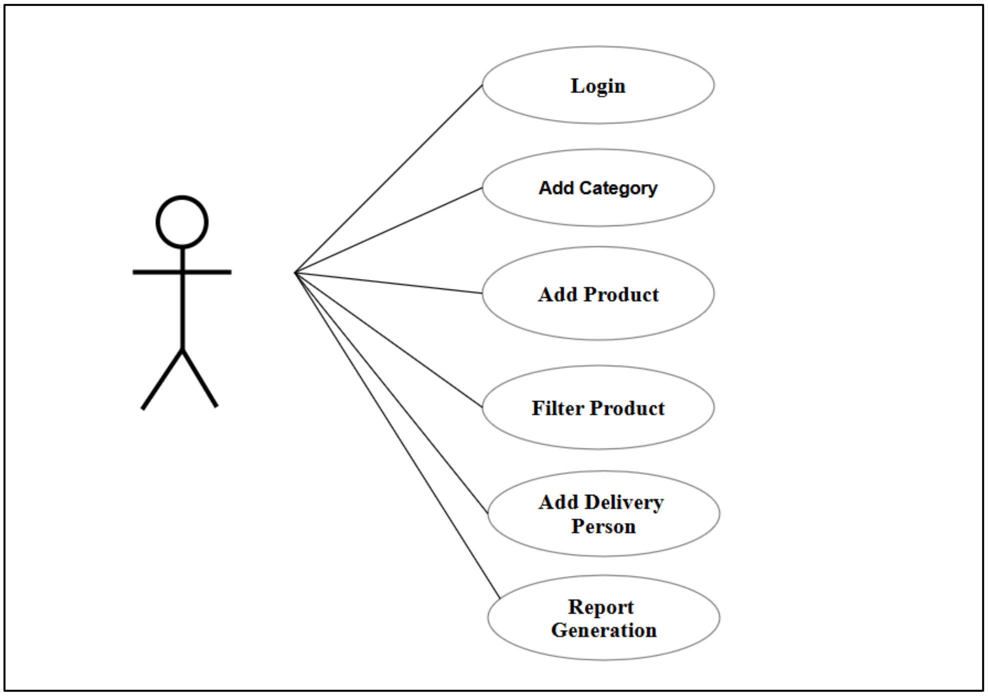
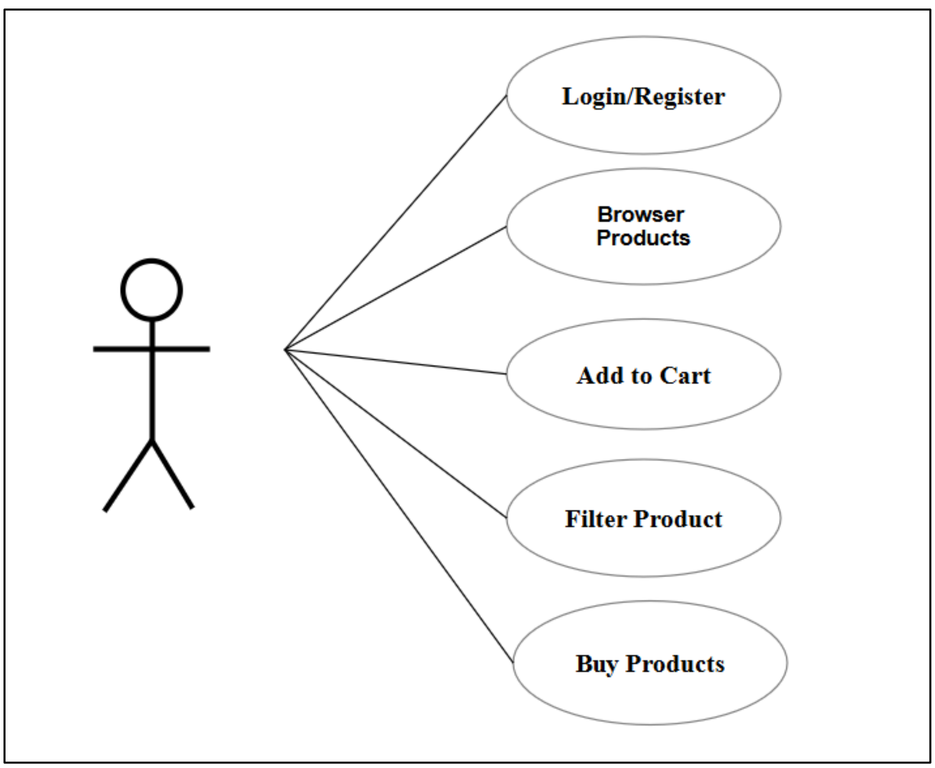
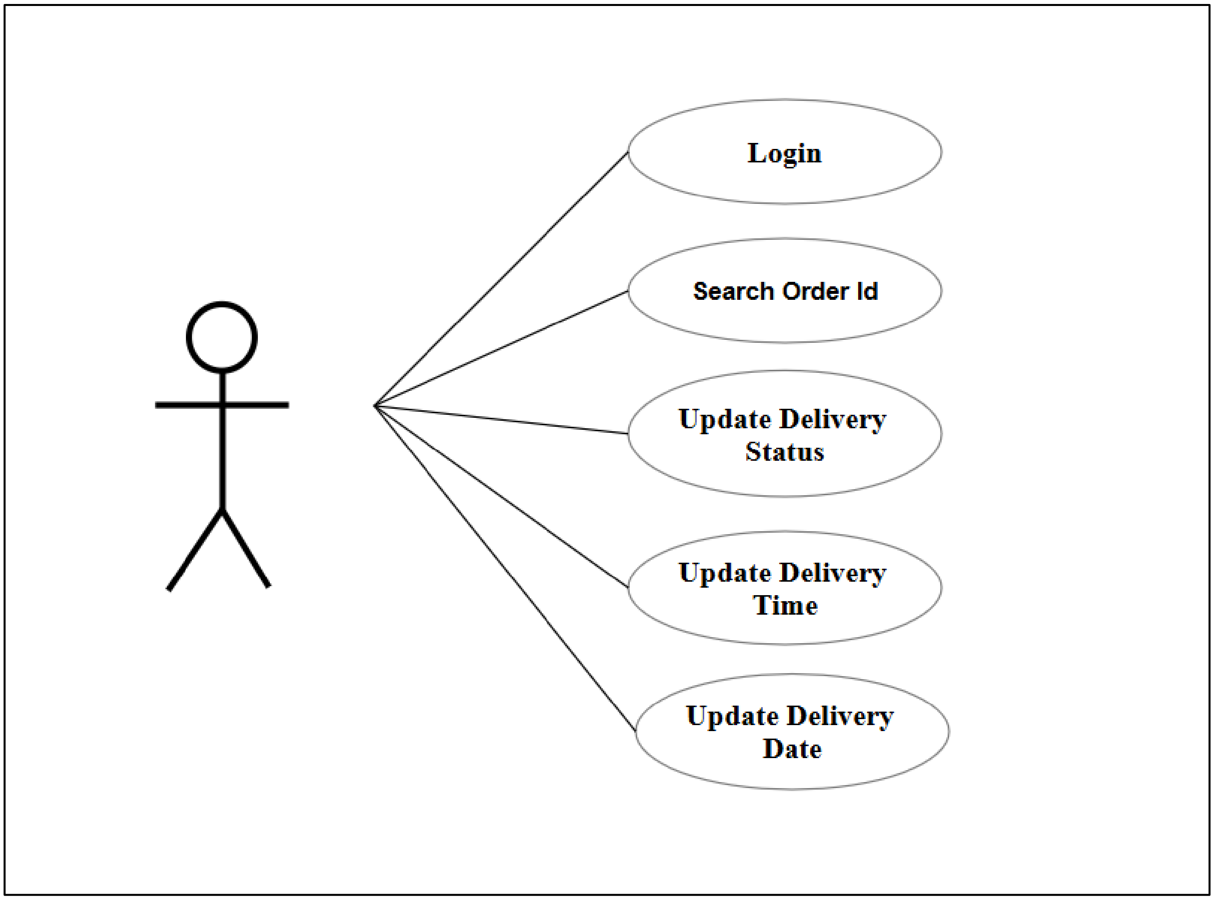


Fig. Use Case Diagram



Customer

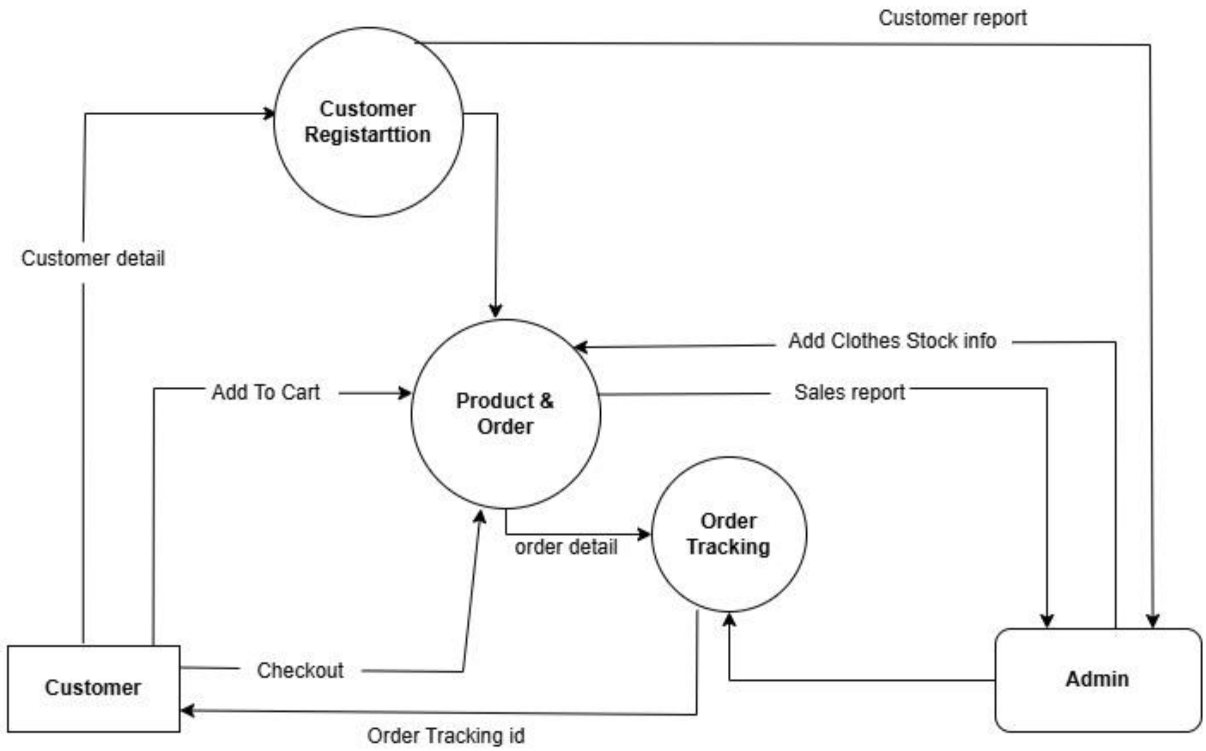


Delivery Person

3.3 Data Flow Diagram:

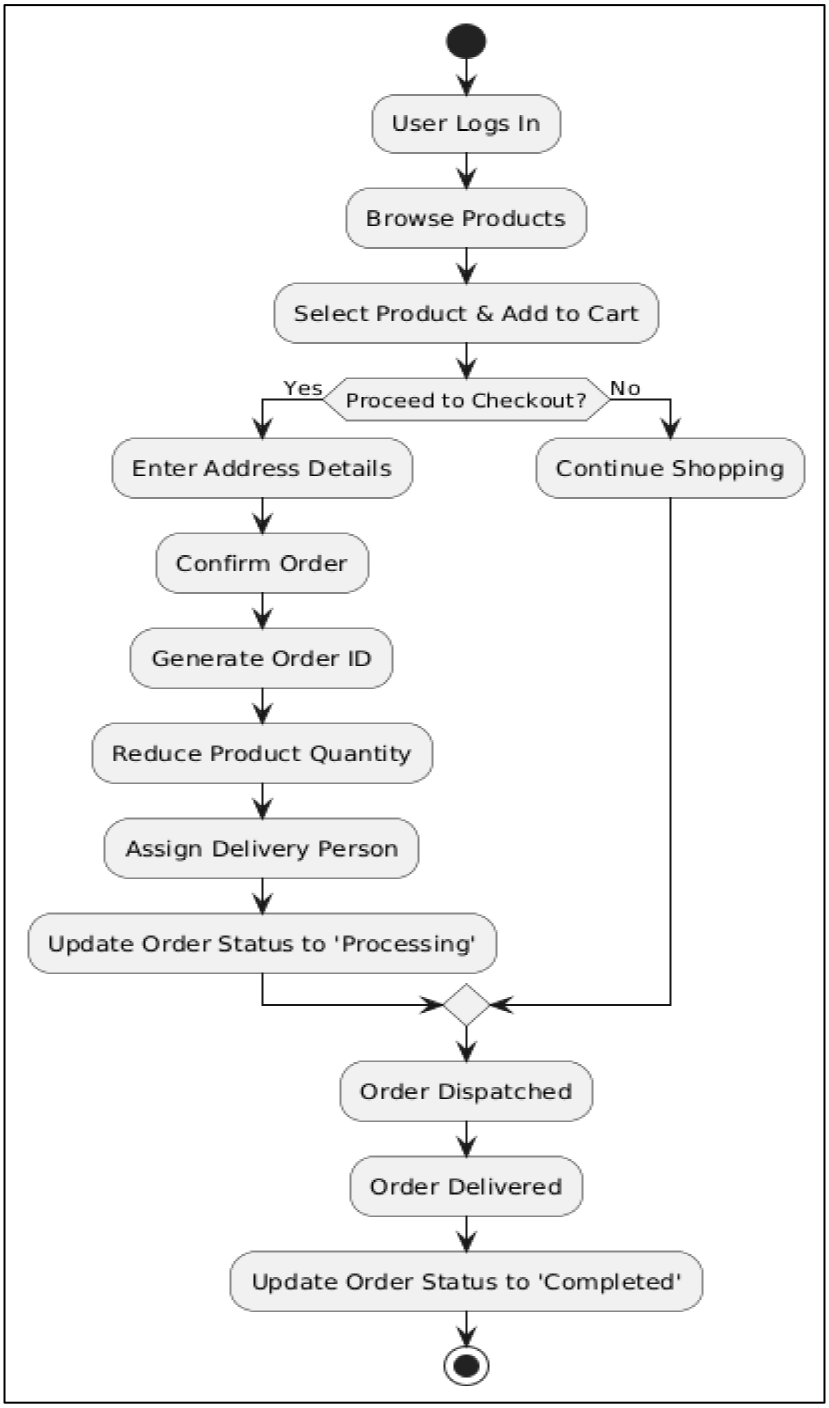
DFD Level 0:

DFD Level 1:

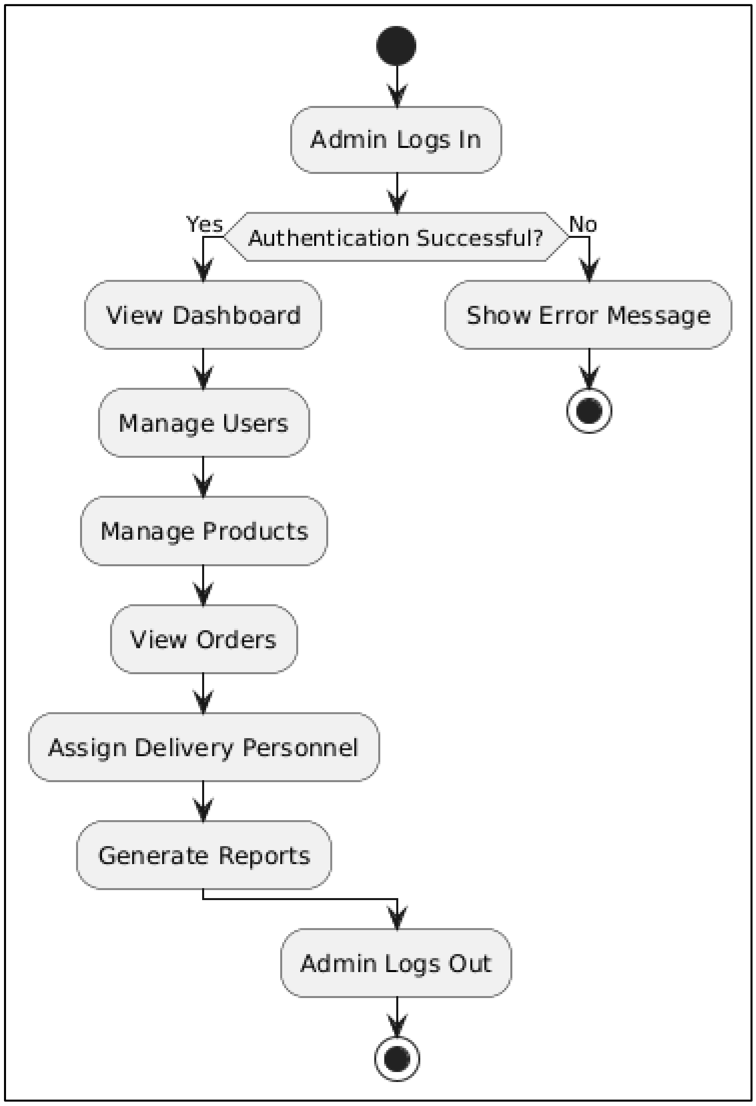

Customer report
Customer Registarttion
Customer detail
Add Clothes Stock info
Add To Cart
Product & Order
Sales report
Order Tracking
order detail
Checkout
Customer
Admin
Order Tracking id

3.4 Activity Diagram :

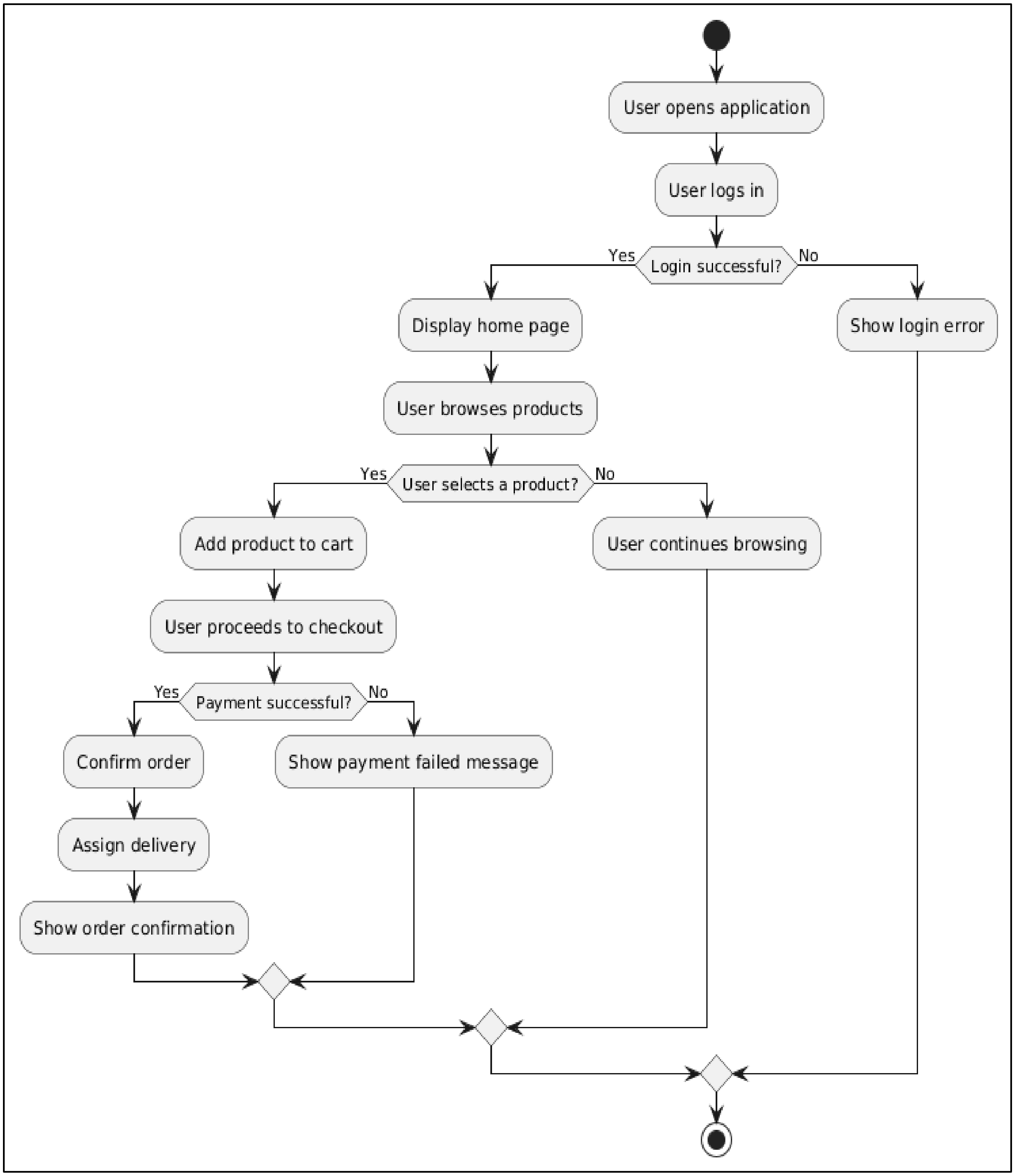
1. Login Activity Diagram



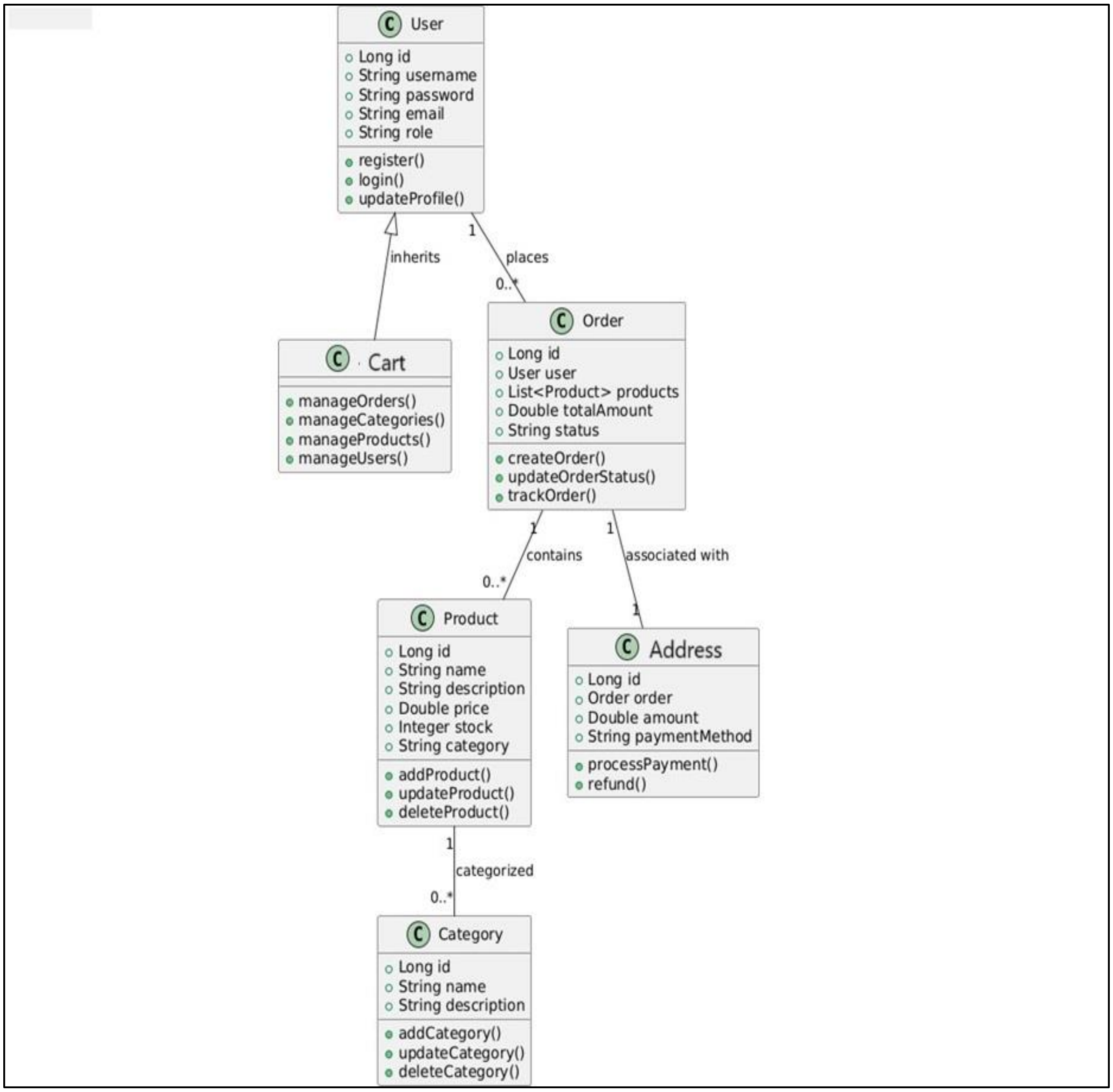
2. Admin Activity Diagram:



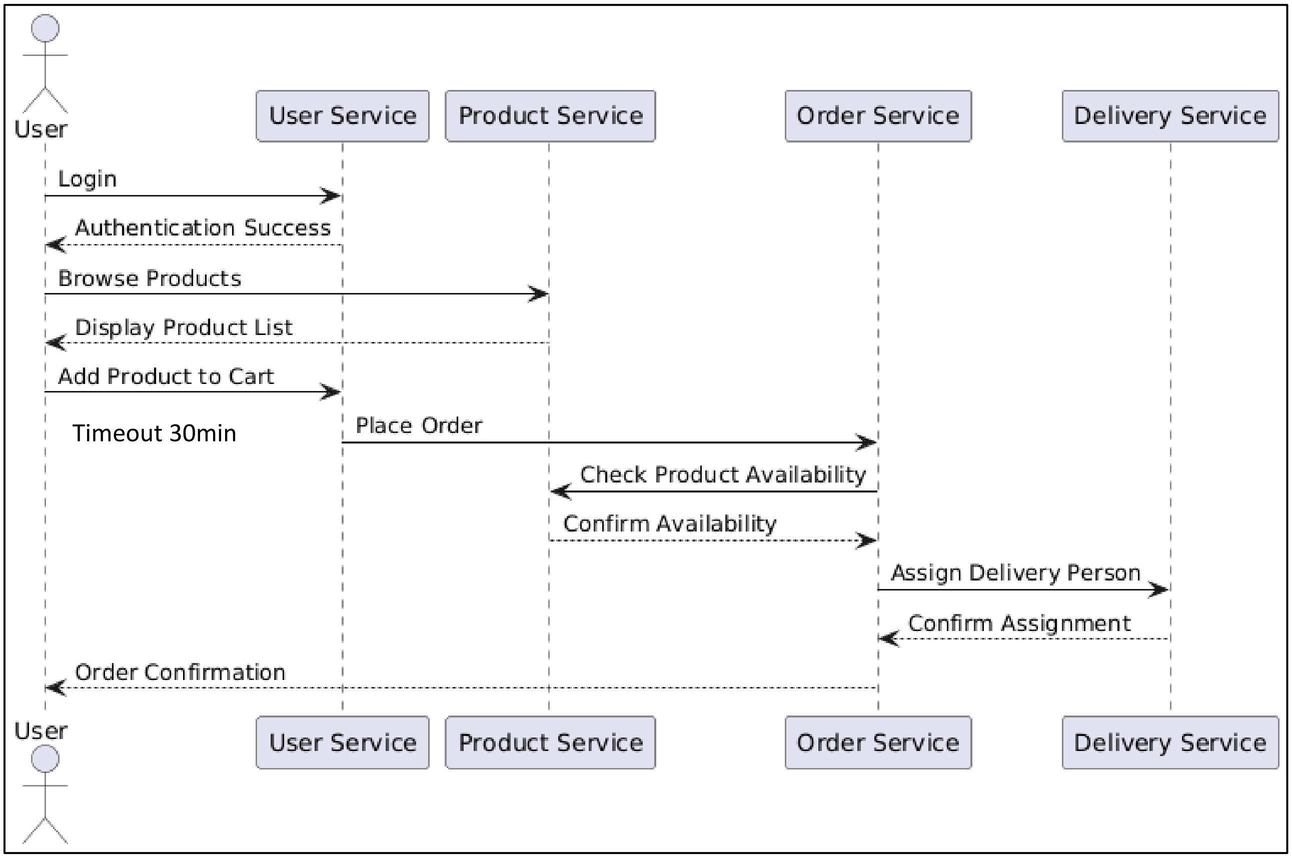
3. User Activity Diagram



3.5 Class Diagram:

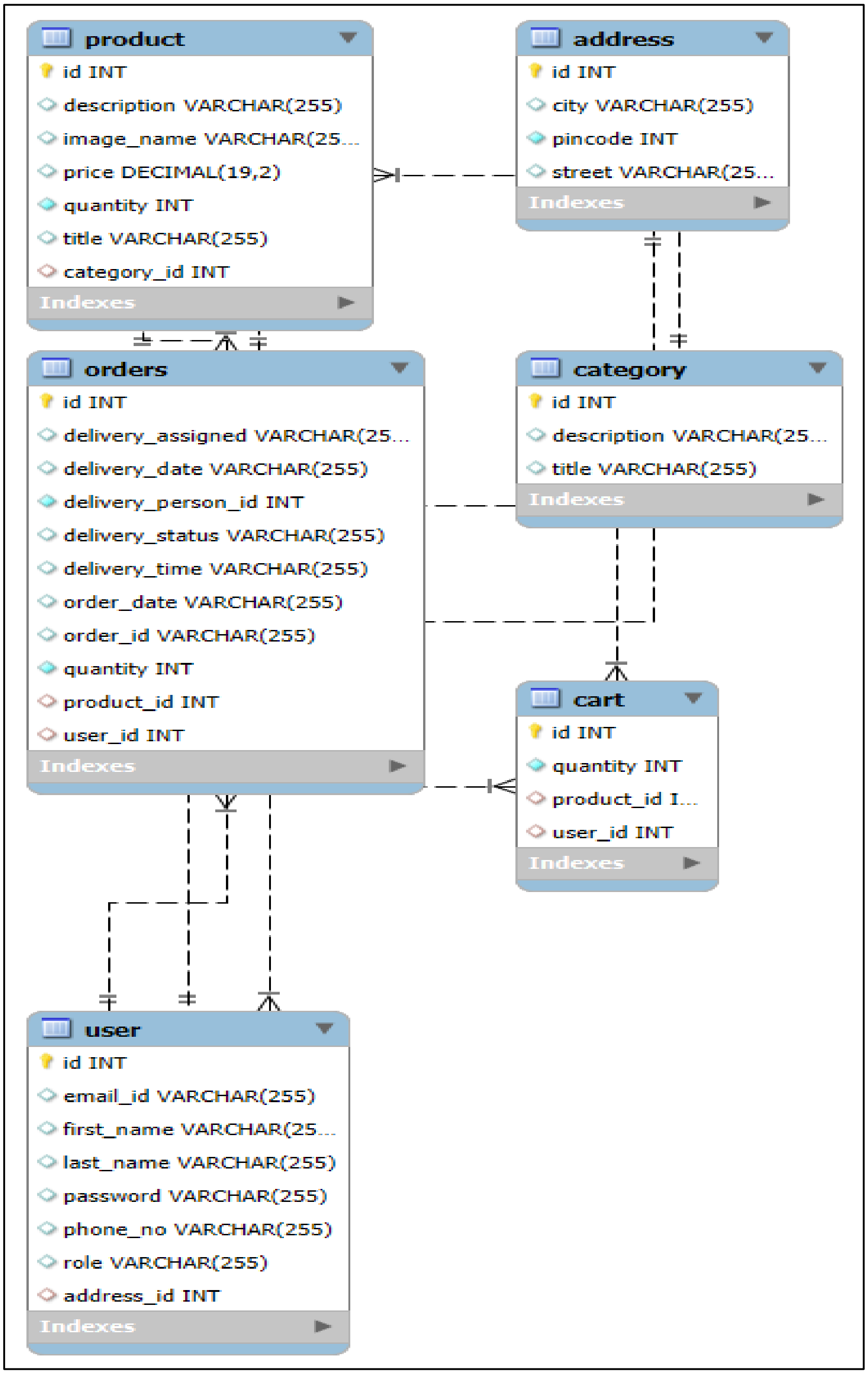


3.6 Sequence Diagram



4. DATABASE DESIGN

4.1 Design:



4.2 Tables:

The following table structures depict the database design.

```
mysql> use online_shopping
Database changed
mysql> show tables;
+---------------------------+
| Tables_in_online_shopping |
+---------------------------+
| address                   |
| cart                      |
| category                  |
| orders                    |
| product                   |
| user                      |
+---------------------------+
6 rows in set (0.11 sec)
```

Database Structures

```
mysql> desc user;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| id         | int          | NO   | PRI | NULL    | auto_increment |
| email_id   | varchar(255) | YES  |     | NULL    |                |
| first_name | varchar(255) | YES  |     | NULL    |                |
| last_name  | varchar(255) | YES  |     | NULL    |                |
| password   | varchar(255) | YES  |     | NULL    |                |
| phone_no   | varchar(255) | YES  |     | NULL    |                |
| role       | varchar(255) | YES  |     | NULL    |                |
| address_id | int          | YES  | MUL | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)
```

Table 1: user

```
mysql> desc orders;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| id                 | int          | NO   | PRI | NULL    | auto_increment |
| delivery_assigned  | varchar(255) | YES  |     | NULL    |                |
| delivery_date      | varchar(255) | YES  |     | NULL    |                |
| delivery_person_id | int          | NO   |     | NULL    |                |
| delivery_status    | varchar(255) | YES  |     | NULL    |                |
| delivery_time      | varchar(255) | YES  |     | NULL    |                |
| order_date         | varchar(255) | YES  |     | NULL    |                |
| order_id           | varchar(255) | YES  |     | NULL    |                |
| quantity           | int          | NO   |     | NULL    |                |
| product_id         | int          | YES  | MUL | NULL    |                |
| user_id            | int          | YES  | MUL | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
11 rows in set (0.00 sec)
```

Table 2: orders

```
mysql> desc product;
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| id          | int           | NO   | PRI | NULL    | auto_increment |
| description | varchar(255)  | YES  |     | NULL    |                |
| image_name  | varchar(255)  | YES  |     | NULL    |                |
| price       | decimal(19,2) | YES  |     | NULL    |                |
| quantity    | int           | NO   |     | NULL    |                |
| title       | varchar(255)  | YES  |     | NULL    |                |
| category_id | int           | YES  | MUL | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

Table 3: Products

mysql> desc category;

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
description	varchar(255)	YES		NULL	
title	varchar(255)	YES		NULL	

3 rows in set (0.00 sec)

Table 4: Categories

mysql> desc cart;

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
quantity	int	NO		NULL	
product_id	int	YES	MUL	NULL	
user_id	int	YES	MUL	NULL	

4 rows in set (0.00 sec)

Table 5 : cart

mysql> desc address;

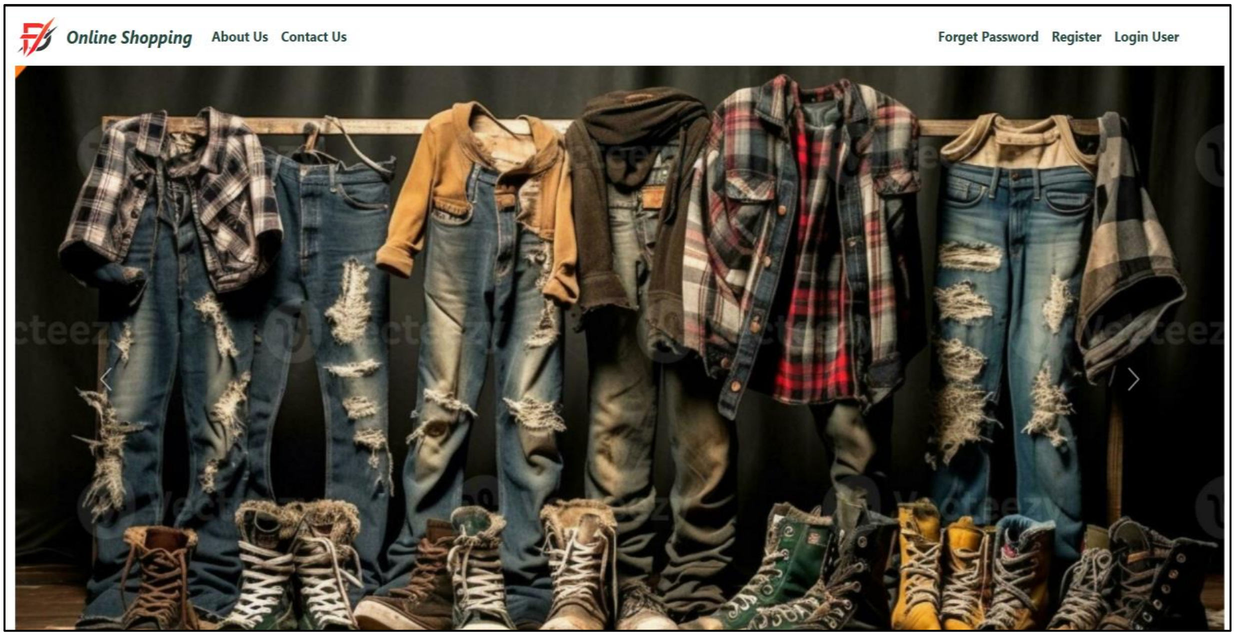
Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
city	varchar(255)	YES		NULL	
pincode	int	NO		NULL	
street	varchar(255)	YES		NULL	

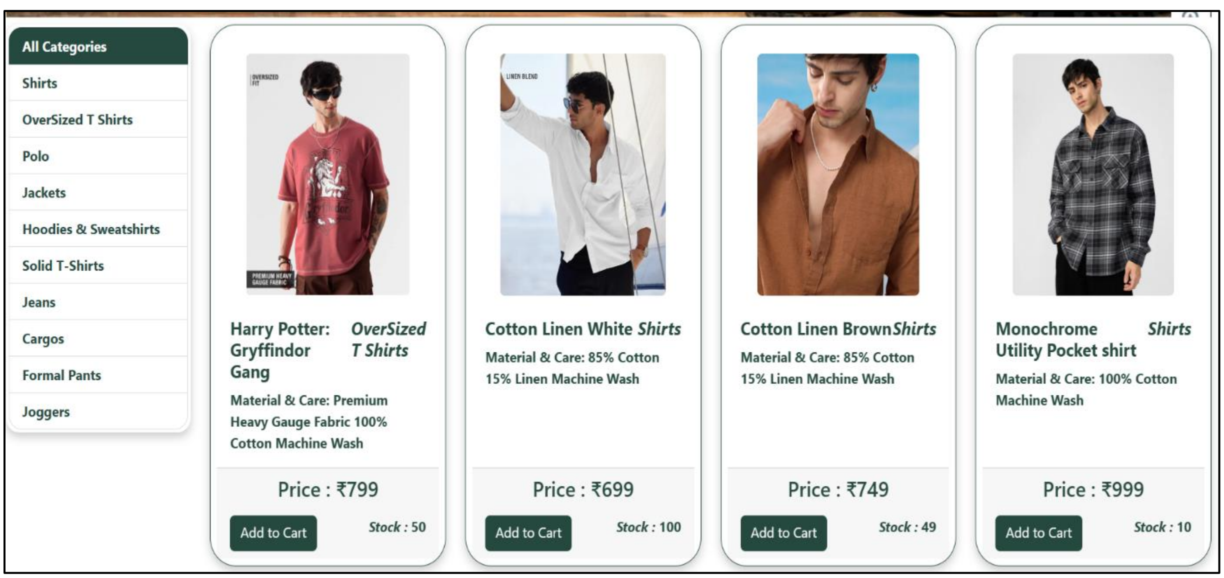
4 rows in set (0.00 sec)

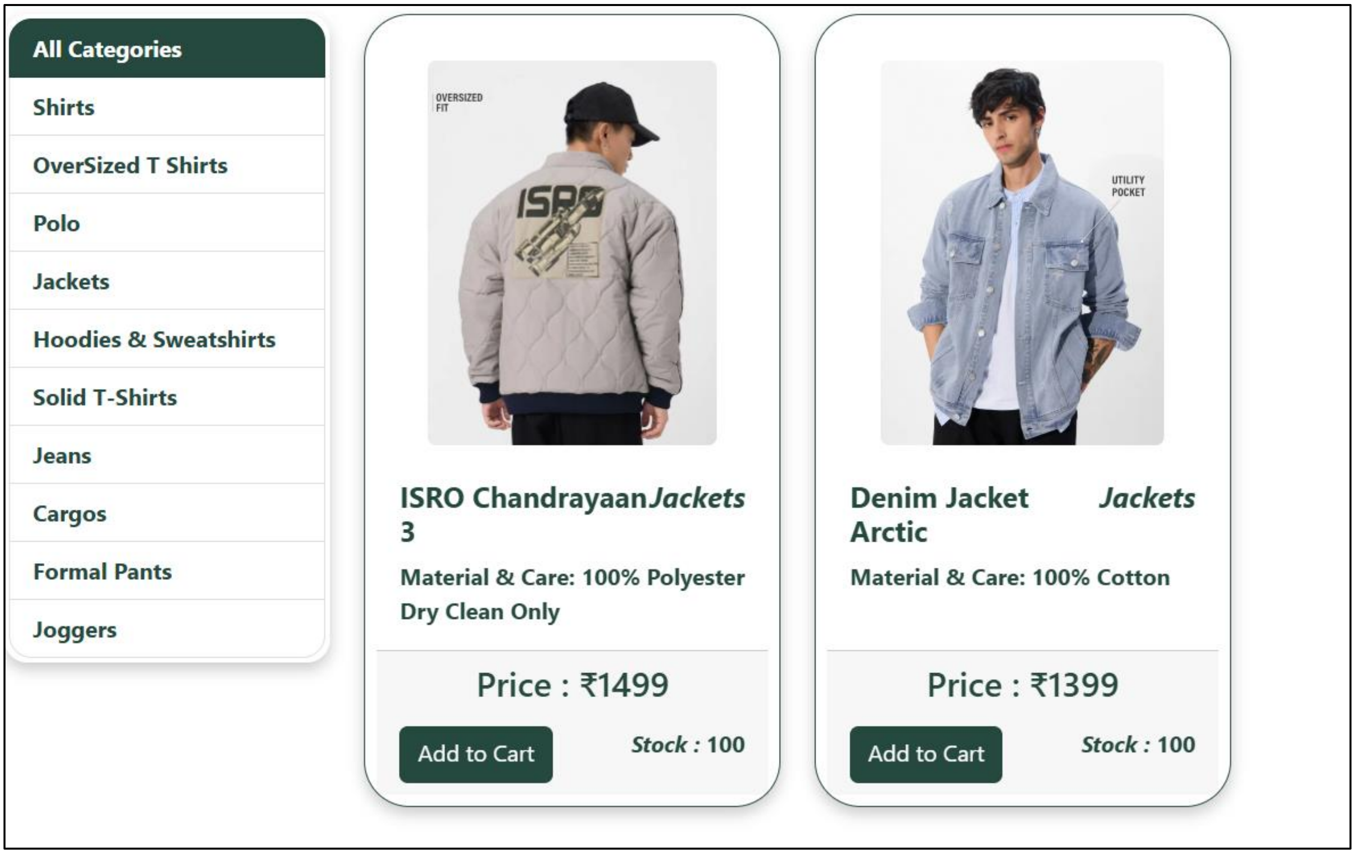
Table5: Address

5 SNAPSHOTS

Home Page:






All Categories
Shirts
OverSized T Shirts
Polo
Jackets
Hoodies & Sweatshirts
Solid T-Shirts
Jeans
Cargos
Formal Pants
Joggers
OVERSIZED FIT
ISRO Chandrayaan 3
Jackets
Material & Care: 100% Polyester Dry Clean Only
Price : ₹1499
Add to Cart
Stock : 100
UTILITY POCKET
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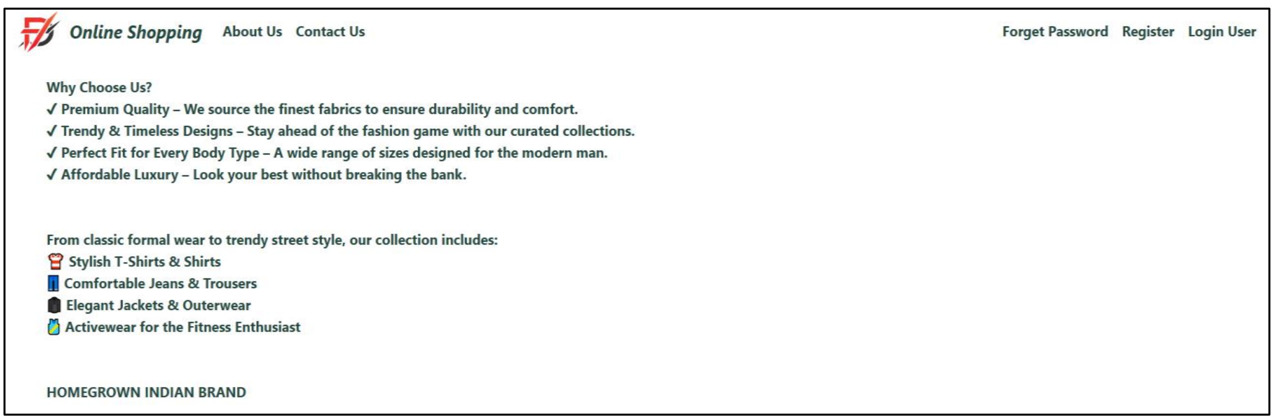
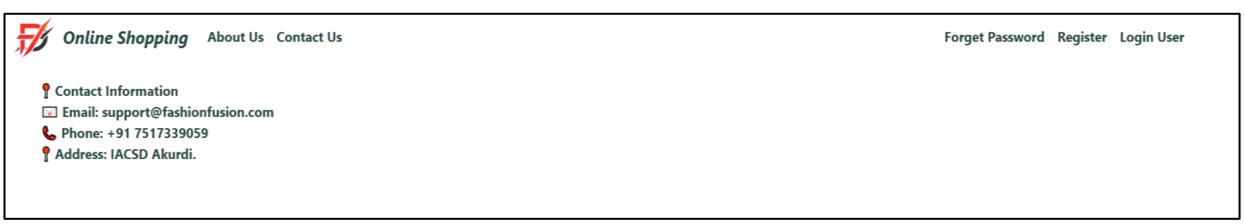
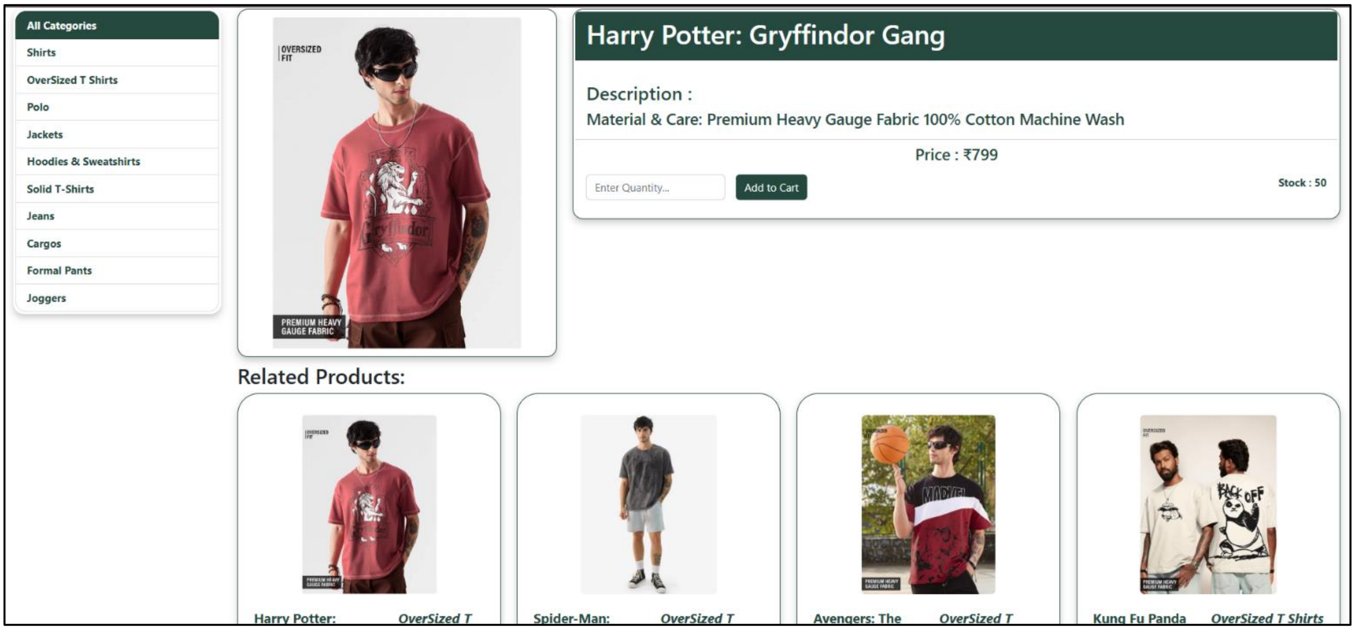
Jackets
Material & Care: 100% Cotton
Price : ₹1399
Add to Cart
Stock : 100


Online Shopping
About Us
Contact Us
Forget Password
Register
Login User
Why Choose Us?
✓ Premium Quality – We source the finest fabrics to ensure durability and comfort.
✓ Trendy & Timeless Designs – Stay ahead of the fashion game with our curated collections.
✓ Perfect Fit for Every Body Type – A wide range of sizes designed for the modern man.
✓ Affordable Luxury – Look your best without breaking the bank.
From classic formal wear to trendy street style, our collection includes:
Stylish T-Shirts & Shirts
Comfortable Jeans & Trousers
Elegant Jackets & Outerwear
Activewear for the Fitness Enthusiast
HOMEGROWN INDIAN BRAND

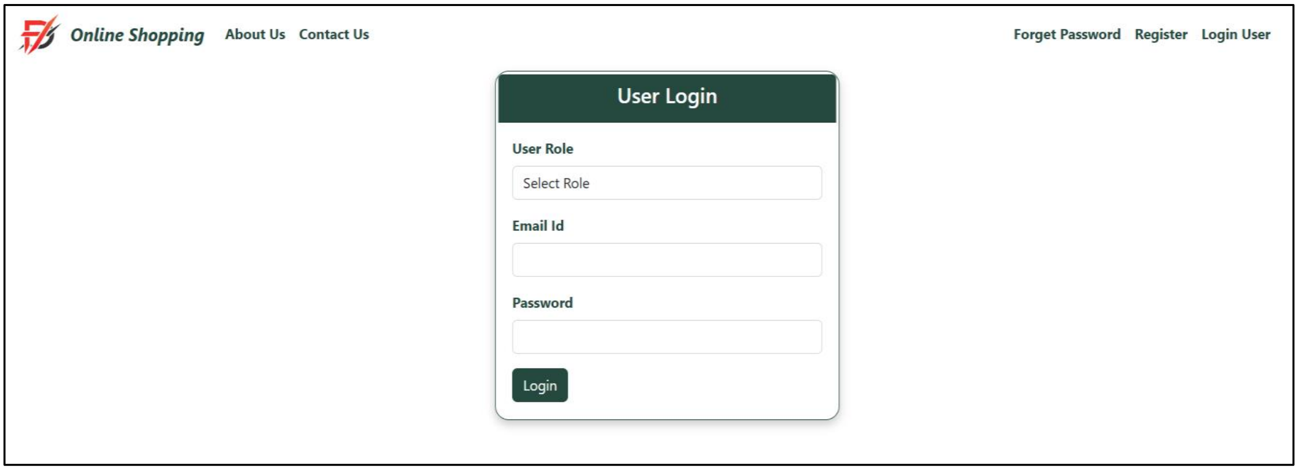

Online Shopping
About Us
Contact Us
Forget Password
Register
Login User
Contact Information
Email: support@fashionfusion.com
Phone: +91 7517339059
Address: IACSD Akurdi.



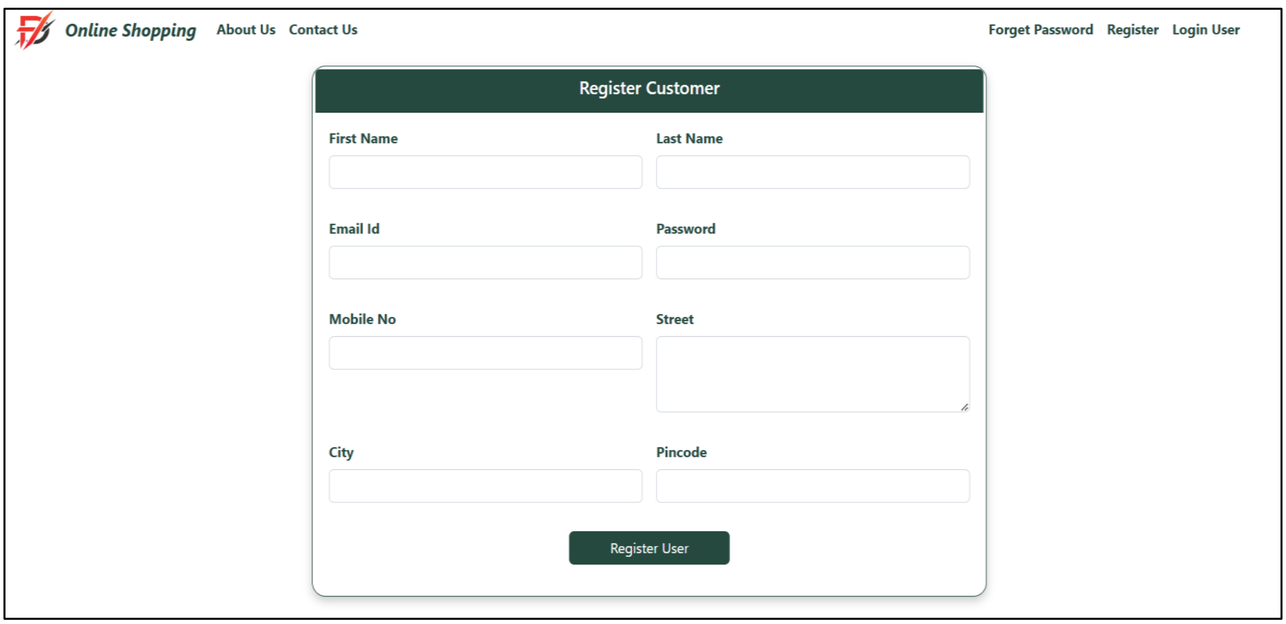
My Orders

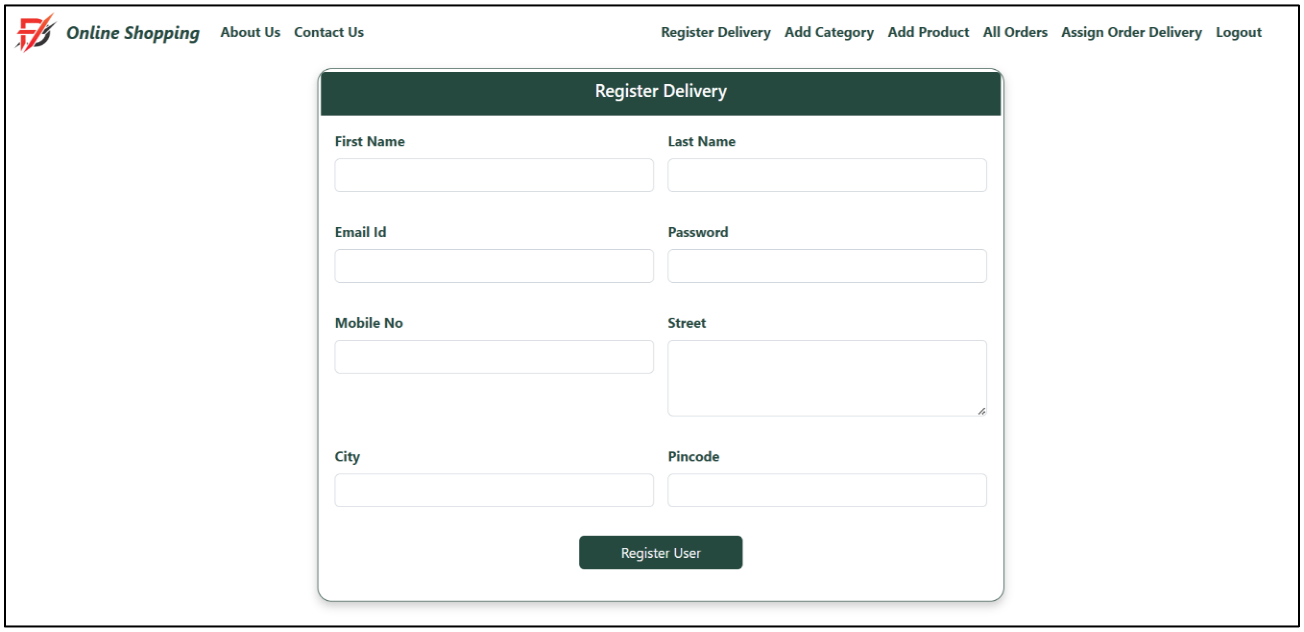
Order Id	Product	Name	Description	Quantity	Total Price	Order Date	Delivery Date	Delivery Status	Delivery Person	Delivery Mobile No
FO0RTLQQFO		Kung Fu Panda	Material & Care: Premium Heavy Gauge Fabric 100% Cotton	1	599.0	10-02-2025 20:40	2025-02-10 Morning	Processing	Ramesh	9876543212
FO0RTLQQFO		Denim Light Blue	98% Cotton 2% Spandex	1	1799.0	10-02-2025 20:40	2025-02-10 Morning	Processing	Ramesh	9876543212
SK2ZLAGL1D		Harry Potter: Gryffindor Gang	Material & Care: Premium Heavy Gauge Fabric 100% Cotton Machine Wash	1	799.0	11-02-2025 00:37	Pending	Pending	Pending	Pending
SK2ZLAGL1D		Aurora Holiday Shirt	Material & Care: 100% Cotton Machine Wash	2	2198.0	11-02-2025 00:37	Pending	Pending	Pending	Pending

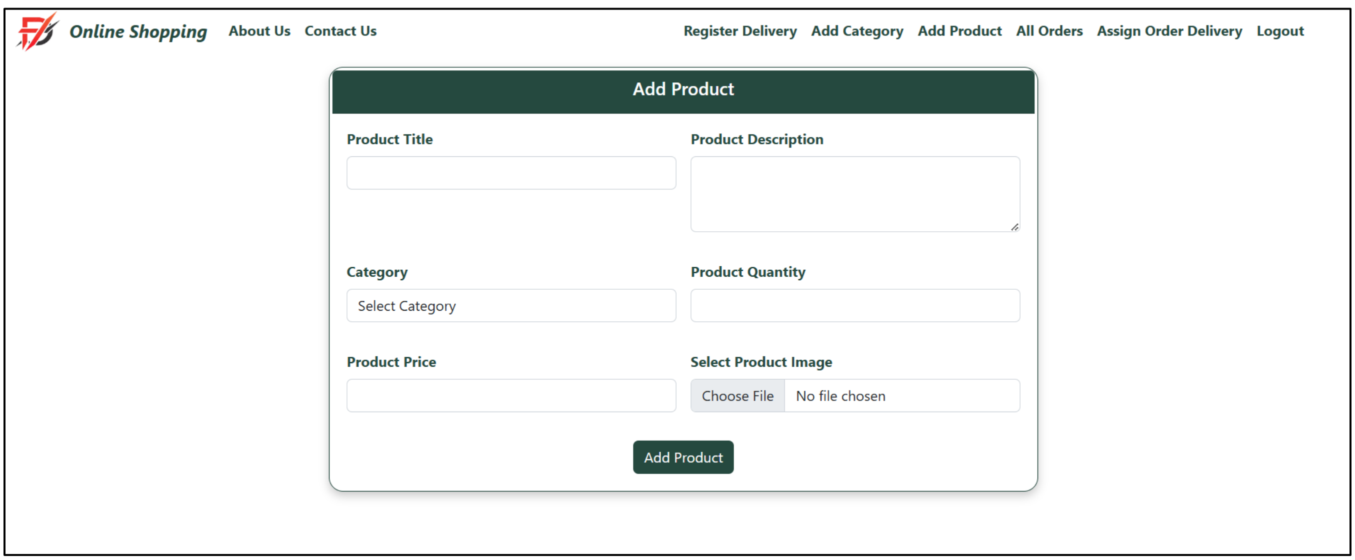
Login :

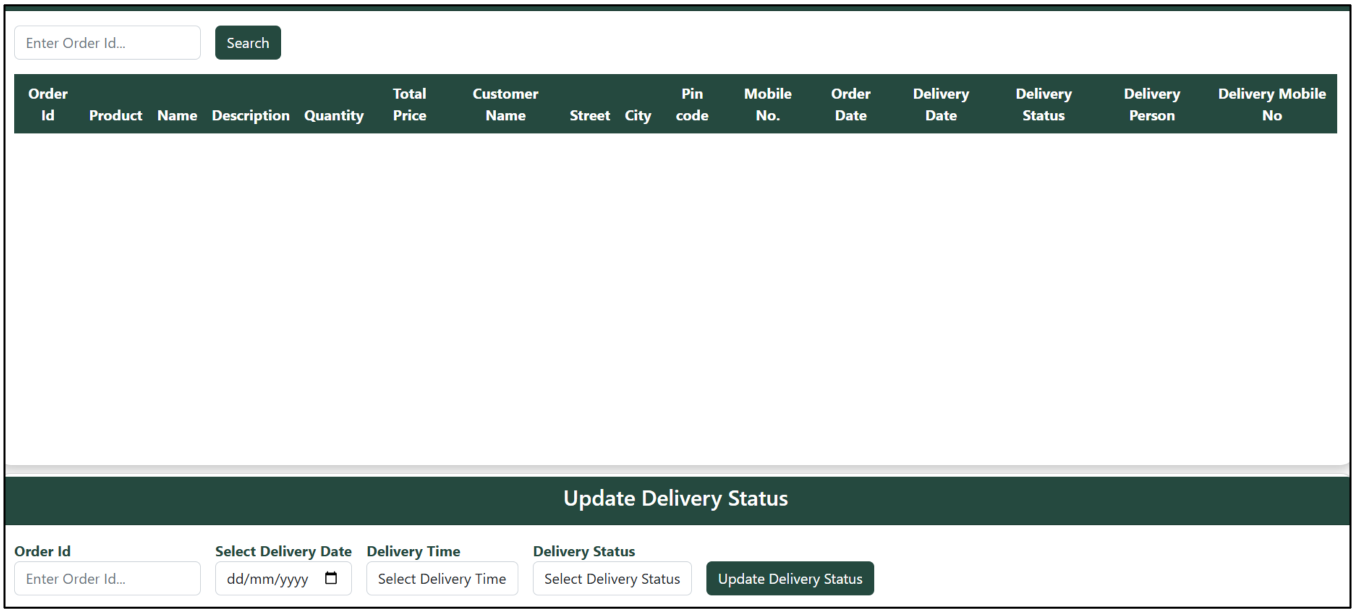


Registration:




Online Shopping
About Us
Contact Us
Register Delivery
Add Category
Add Product
All Orders
Assign Order Delivery
Logout
Register Delivery
First Name
Last Name
Email Id
Password
Mobile No
Street
City
Pincode
Register User


Online Shopping
About Us
Contact Us
Register Delivery
Add Category
Add Product
All Orders
Assign Order Delivery
Logout
Add Product
Product Title
Product Description
Category
Select Category
Product Quantity
Product Price
Select Product Image
Choose File
No file chosen
Add Product


Enter Order Id...
Search
Order Id
Product
Name
Description
Quantity
Total Price
Customer Name
Street
City
Pin code
Mobile No.
Order Date
Delivery Date
Delivery Status
Delivery Person
Delivery Mobile No
Update Delivery Status
Order Id
Enter Order Id...
Select Delivery Date
dd/mm/yyyy
Delivery Time
Select Delivery Time
Delivery Status
Select Delivery Status
Update Delivery Status

6 CONCLUSION

The development of the Men’s Clothing E-Commerce Website using Spring Boot, Spring Security, React, and MySQL showcases the power of modern web technologies in creating a secure, scalable, and user-friendly online shopping platform. This project successfully integrates a dynamic frontend with React-Redux, ensuring a smooth and responsive user experience, while the backend, powered by Spring Boot, provides robust functionality for user authentication, product management, order processing, and secure transactions.

By implementing Spring Security, the platform ensures data protection and user authentication through role-based access control (RBAC). Additionally, MySQL efficiently manages customer, product, and order data, ensuring seamless operations. The integration of key e-commerce functionalities, such as product browsing, cart management, secure checkout, and order tracking, enhances the overall shopping experience.

This project demonstrates the successful combination of modern web frameworks, backend services, and database management to build a full-stack e-commerce platform. Future improvements could include AI-driven product recommendations, advanced analytics, and multi-vendor support to further enhance the platform’s functionality and scalability.

Overall, this Men’s Clothing E-Commerce Website serves as a practical and secure solution for online fashion retail, offering a reliable and engaging shopping experience for customers.

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